

mtDNA-FM: A Domain-Specialized Foundation Model for Mitochondrial DNA with Circular Positional Encoding

Thawfeek Varusai¹

¹Independent Researcher

Correspondence: vthawfeek@gmail.com

ORCID: 0000-0002-7864-5971

Abstract

Mitochondrial DNA (mtDNA) is a 16,569 bp circular genome with central roles in human disease, population genetics, and evolutionary biology. Existing DNA foundation models are pre-trained on nuclear genomes and use linear positional encodings, which misrepresent the circular topology of mtDNA: positions 0 and 16,568 are physically adjacent at the D-loop junction but receive maximally different encodings.

We present **mtDNA-FM**, the first foundation model dedicated to mitochondrial DNA. It introduces two domain-specific design choices: (i) a *circular positional encoding* where positions 0 and 16,568 receive near-identical encodings ($\epsilon \approx 3.8 \times 10^{-4}$ per dimension), correctly representing their adjacency at the D-loop; and (ii) a *heteroplasmy projection channel* that accepts a continuous per-position variant allele fraction alongside sequence content. The model is pre-trained on 152,590 vertebrate mtDNA sequences in two phases (117,615 cross-species, then 34,975 human HmtDB).

Zero-shot evaluation of the Phase 1 checkpoint achieves 37.9% on 26-class haplogroup classification (95% CI 34.4–41.2%; random baseline 3.85%; $9.9\times$ lift) and AUROC 0.777 (95% CI 0.731–0.821) on pathogenic variant discrimination without labels. Comparison with DNABERT-2 (66.3%, 117M parameters) reveals a clade-specific gap in West Eurasian haplogroups. Per-class analysis yields four observations that identify specific representation failures and map concrete improvements for a next-generation model.

Availability: <https://github.com/vthawfeek/mtdna-foundation-model>, <https://huggingface.co/vthawfeek/mtdna-foundation-model>, <https://huggingface.co/spaces/vthawfeek/mtdna-fm-demo>

1 Introduction

Mitochondrial DNA (mtDNA) is a 16,569 bp circular, double-stranded genome present in hundreds to thousands of copies per cell [Anderson et al., 1981]. mtDNA is maternally inherited without recombination, making it a key tool for phylogenetics, population genetics, and forensic identification [van Oven and Kayser, 2009]. Clinically, pathogenic mtDNA variants cause severe multi-system disorders including MELAS, LHON, and Leigh syndrome, collectively affecting at least 1 in 5,000 individuals [DiMauro and Schon, 2003]. A further complication is *heteroplasmy*: the co-existence of wild-type and mutant mtDNA molecules in the same cell at varying proportions. Disease penetrance for many pathogenic variants depends critically on the heteroplasmy level [Stewart and Chinnery, 2015], yet predicting this from sequence context remains an open problem.

DNABERT-2 [Zhou et al., 2024], HyenaDNA [Nguyen et al., 2023], the Nucleotide Transformer [Dalla-Torre et al., 2023], and Evo [Nguyen et al., 2024] show that pre-training on DNA sequences yields representations that transfer across genomic tasks. All four are pre-trained on nuclear or multi-kingdom genomes without a dedicated mitochondrial corpus or mtDNA-specific design objectives, and share two design gaps for mtDNA:

1. **Linear positional encoding.** Standard sinusoidal PE [Vaswani et al., 2017] treats positions 0 and 16,568 as maximally distant. In the circular mtDNA molecule, these positions are physically adjacent

at the D-loop junction. This creates incorrect representations of any functional element spanning the circular boundary.

2. **No heteroplasmy channel.** Existing sequence models assume one definitive base per position. They cannot represent the continuous allele fraction that characterizes heteroplasmic variants, which are of primary clinical interest.

Nuclear DNA also contains repeat elements, splice sites, CpG islands, and regulatory grammar absent from mtDNA. Representations built on billions of bases of nuclear sequence may not transfer to a 16,569 bp genome with different base composition, gene density, and evolutionary constraint.

We present **mtDNA-FM**, a BERT-style [Devlin et al., 2019] foundation model designed and pre-trained for mitochondrial DNA. Our contributions are:

1. A **circular positional encoding** stored as a non-learnable buffer (Section 3.1). Positions 0 and 16,568 are physically adjacent at the D-loop boundary but receive maximally different standard sinusoidal encodings. Our circular encoding eliminates this mismatch: the two positions receive near-identical encodings ($\varepsilon \approx 3.8 \times 10^{-4}$ per dimension).
2. A **heteroplasmy projection channel** that accepts per-position variant allele fractions as a continuous input alongside sequence content, enabling the model to represent heteroplasmic variants (Section 3.1).
3. A **two-phase domain-adaptive curriculum**: Phase 1 on 117,615 cross-species vertebrate mtDNA sequences to learn conserved structural features, followed by Phase 2 on 34,975 human HmtDB sequences with heteroplasmy loss enabled (Section 3.2).
4. **Zero-shot evaluation** on 26-class haplogroup classification and pathogenic variant discrimination, with per-class analysis against DNABERT-2 (Section 4).

All evaluations in this preprint use the Phase 1 checkpoint. Ablations of the circular PE, heteroplasmy channel, and Phase 1 vs. Phase 2 curriculum are deferred to the extended paper.

2 Related Work

2.1 Genomic Foundation Models

DNABERT [Ji et al., 2021] was the first BERT-based model for DNA, pre-trained on the human reference genome with 6-mer tokenization. DNABERT-2 [Zhou et al., 2024] extended this to genomes from 32 multi-kingdom species with BPE tokenization and achieved state-of-the-art on the Genome Understanding Evaluation (GUE) benchmark. The Nucleotide Transformer [Dalla-Torre et al., 2023] scales to 2.5B parameters across 850 genomes. HyenaDNA [Nguyen et al., 2023] uses the Hyena implicit convolution operator for million-base context windows. Evo [Nguyen et al., 2024] scales to 7B parameters on prokaryotic and eukaryotic sequences and supports sequence generation and editing. All these models use linear positional encoding, have no heteroplasmy modeling, and were not pre-trained with a dedicated mitochondrial corpus or designed for circular mtDNA topology.

2.2 mtDNA Bioinformatics Tools

Classical mtDNA analysis uses specialized tools: HaploGrep 2 [Weissensteiner et al., 2016] classifies haplogroups from variant calls using PhyloTree [van Oven and Kayser, 2009] branch definitions; Haplocheck [Weissensteiner et al., 2021] detects contamination and heteroplasmy from sequencing data. Pathogenicity databases include MITOMAP [Lott et al., 2013], ClinVar [Landrum et al., 2018], and HelixMTdb [Bolze et al., 2022]. MitImpact [Castellana and Mazza, 2013] aggregates computational pathogenicity scores for protein-coding variants. None of these tools use pre-trained deep learning representations.

2.3 Variant Effect Prediction

Classical methods (SIFT [Ng and Henikoff, 2003], PolyPhen-2 [Adzhubei et al., 2010]) score variants using evolutionary conservation from multiple sequence alignments. Deep generative models (EVE [Brandes et al., 2023], ESM-1v [Meier et al., 2021]) show that pre-trained sequence distributions capture functional constraint. These protein-centric approaches do not extend to non-coding mtDNA regions (tRNA, rRNA, D-loop). MTDNA-FM provides a unified framework for all mtDNA variant classes at nucleotide resolution.

2.4 Positional Encoding Variants

Standard sinusoidal PE [Vaswani et al., 2017] and its learnable counterpart treat position as a linear index. Relative PE methods (RoPE [Su et al., 2024], ALiBi [Press et al., 2022]) encode pairwise distance between tokens, improving length generalization but not circular topology. Sinusoidal and rotary encodings are periodic in phase, but their period is not aligned to any physical sequence length, so a sequence’s two endpoints do not become adjacent. To our knowledge, no prior genomic foundation model aligns the positional period to the length of a circular genome so that its two endpoints receive near-identical encodings; this alignment to circular topology is the specific contribution we claim.

3 Methods

3.1 Architecture

MTDNA-FM is a BERT-style [Devlin et al., 2019] transformer encoder with 6 layers, 8 attention heads, 256-dimensional hidden states, and 1,024-dimensional feed-forward layers (~ 5.8 M parameters). It uses pre-layer normalisation for training stability [Xiong et al., 2020]. Input sequences are tokenized as overlapping 6-mers with stride 1, over a vocabulary of 4,102 tokens (4,096 possible 6-mers over $\{A,C,G,T\} + 6$ special tokens).

3.1.1 Circular Positional Encoding

Standard sinusoidal PE assigns position p an angle proportional to $p/\text{max_len}$. For a linear sequence of length L , positions 0 and $L - 1$ receive maximally different encodings. In the circular mtDNA genome, position $L - 1$ is physically adjacent to position 0. We resolve this with a circular angle:

$$\phi_p = \frac{2\pi p}{L_{\text{genome}}} \quad (1)$$

where $L_{\text{genome}} = 16,569$ bp is the rCRS genome length. The positional encoding is then:

$$\text{PE}(p, 2i) = \sin\left(\frac{\phi_p}{10000^{2i/d}}\right) \quad (2)$$

$$\text{PE}(p, 2i + 1) = \cos\left(\frac{\phi_p}{10000^{2i/d}}\right) \quad (3)$$

where $d = 256$ is the hidden dimension. The buffer stores positions $p \in \{0, \dots, L_{\text{genome}} - 1\}$. At the last position $p = L_{\text{genome}} - 1 = 16,568$, the angle is $\phi_{16568} = 2\pi \times 16568/16569 \approx 2\pi - 3.8 \times 10^{-4}$, giving $|\text{PE}(16568) - \text{PE}(0)| \approx 3.8 \times 10^{-4}$ per dimension. Positions 0 and 16,568 therefore receive *approximately* identical encodings, with the angular gap equal to the spacing between any two adjacent positions ($\Delta\phi = 2\pi/16569$). The encoding is stored as a non-learnable `nn.Buffer`.

3.1.2 Heteroplasmy Projection Channel

Each input position receives an optional continuous scalar $h_p \in [0, 1]$ representing the variant allele fraction. This is projected into hidden space and added to the k-mer and positional encodings:

$$\mathbf{e}_p = \text{Embed}_{\text{kmer}}(x_p) + \text{PE}(p) + \text{LayerNorm}(\mathbf{W}_{\text{het}} h_p + \mathbf{b}_{\text{het}}) \quad (4)$$

Figure 1: mtDNA-FM — Circular Genome Representation and Architecture

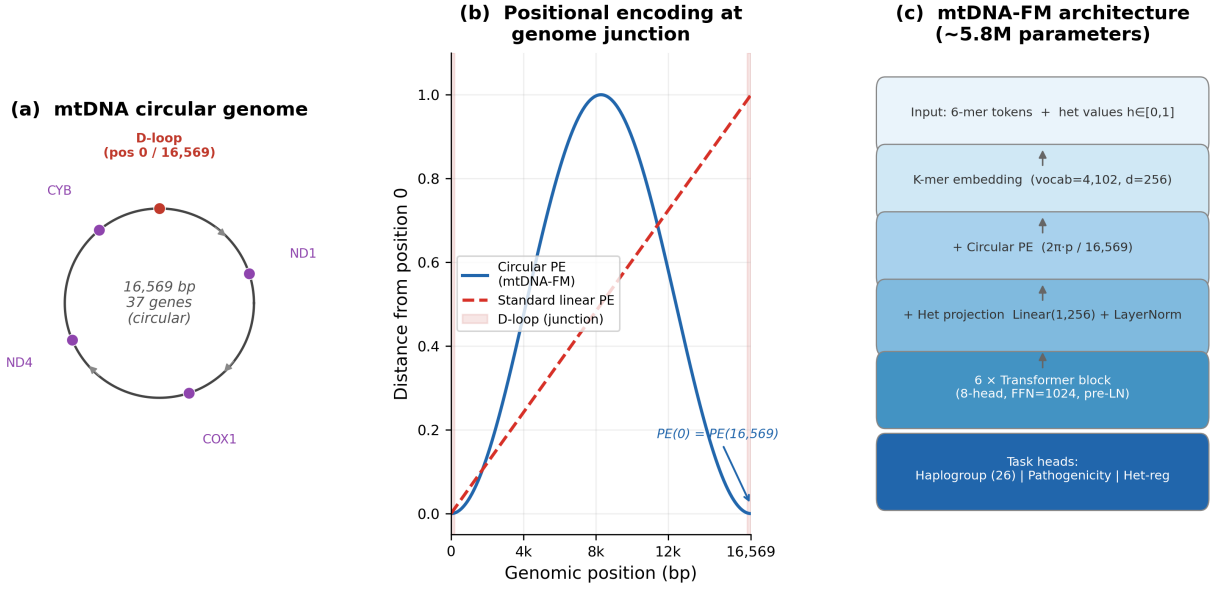


Figure 1: **mtDNA-FM architecture**. (a) The 16,569 bp circular mitochondrial genome with key gene landmarks. (b) Positional distance from position 0 for standard linear PE vs. circular PE. The circular encoding returns to approximately zero at position 16,568 (the last genomic position), minimising the artificial positional discontinuity at the D-loop junction. (c) Model block diagram showing the three input channels (k-mer embedding, circular PE, heteroplasmy projection) feeding into 6 transformer layers.

where $\mathbf{W}_{\text{het}} \in \mathbb{R}^d$ and $\mathbf{b}_{\text{het}} \in \mathbb{R}^d$ are the learnable weight and bias of an `nn.Linear(1, d)` projection. LayerNorm normalizes the projection output to match the scale of the k-mer embedding. The bias term $\mathbf{b}_{\text{het}}$ is necessary for the channel to encode different heteroplasmy levels: `LayerNorm( $\mathbf{W}_{\text{het}} h_p$ )` without a bias is invariant to the scale of h_p for any $h_p > 0$, reducing the channel to a binary (present/absent) indicator; the bias $\mathbf{b}_{\text{het}}$ breaks this scale invariance so that distinct values of h_p produce distinct embeddings. When $h_p = 0$ for all positions (Phase 1, where no per-position heteroplasmy targets are available), the channel outputs `LayerNorm( $\mathbf{b}_{\text{het}}$ )`, a small constant offset relative to the k-mer and positional embeddings.

3.2 Pre-training

3.2.1 Datasets

Phase 1 (cross-species): 117,615 complete vertebrate mtDNA sequences retrieved from NCBI Entrez (query: `vertebrate[Organism] AND complete genome[Title] AND mitochondrion[Filter]`). Sequences span mammals, birds, reptiles, amphibians, and fish.

Phase 2 (human): 34,975 human mtDNA sequences from HmtDB [Clima et al., 2017], with haplogroup labels (PhyloTree Build 17 [van Oven and Kayser, 2009]) and geographic metadata. Sequences not exactly 16,569 bp in length were length-normalized before tokenization: sequences shorter than 16,569 bp had N characters inserted at the D-loop start position (rCRS position 576), preserving protein-coding gene coordinates; sequences longer than 16,569 bp were trimmed from the 3' end. A quality filter retaining sequences with at most 10% ambiguous bases after normalization keeps 99.93% of sequences.

3.2.2 Training Objective

Phase 1 uses masked language modeling (MLM; Devlin et al. 2019): 15% of tokens are masked using the 80/10/10 scheme (80% [MASK], 10% random, 10% unchanged). The D-loop homopolymeric C-tract specifically (positions 303–315, 12 positions out of the 576 bp D-loop) is excluded from masking due to high sequencing

Figure 2: Positional similarity matrix — standard vs circular positional encoding
Red dashed lines mark the D-loop region (positions 0 and ~16,024-16,569)

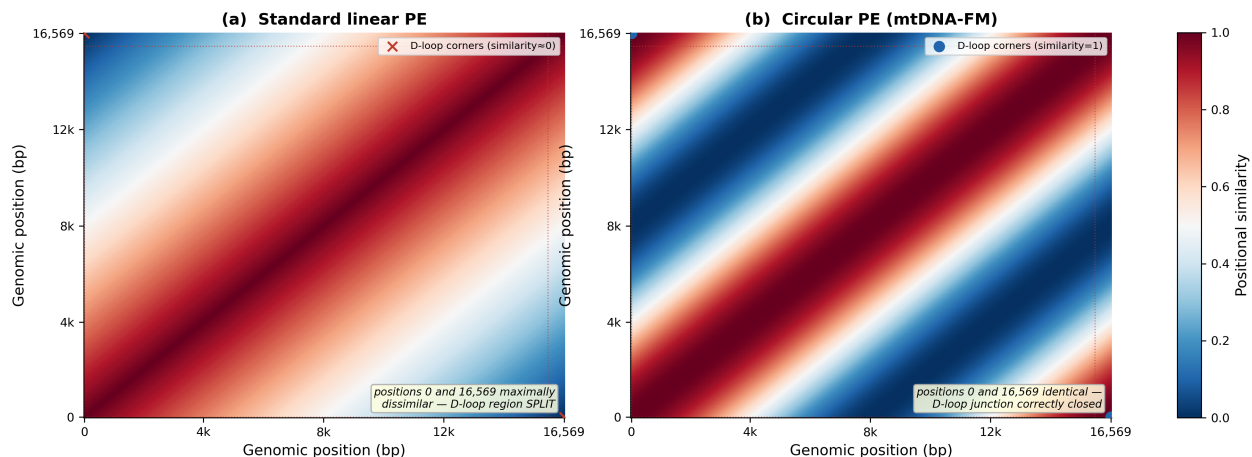


Figure 2: **Positional similarity matrices for standard vs. circular positional encoding.** Each cell (i, j) shows positional similarity between positions i and j . (a) Standard sinusoidal PE: positions 0 and 16,568 are maximally dissimilar, splitting the D-loop across an artificial boundary. (b) Circular PE: positions 0 and 16,568 receive approximately identical encodings ($\epsilon \approx 3.8 \times 10^{-4}$ per dimension), correctly representing their adjacency. Red dashed lines mark the D-loop region.

error rates at homopolymeric runs; the remainder of the D-loop is included in the masked training objective. Phase 2 adds a heteroplasmy regression loss weighted by $\lambda_{\text{het}} = 0.3$:

$$\mathcal{L} = \mathcal{L}_{\text{MLM}} + \lambda_{\text{het}} \mathcal{L}_{\text{het}} \quad (5)$$

Heteroplasmy regression targets are derived from the gnomAD v3.1 chrM variant table: for each position with a variant entry, y_{het} is set to the `mean_h1` INFO field (mean variant allele fraction across heteroplasmic carriers at that site), using only variants with at least 50 heteroplasmic carriers (`n_het` ≥ 50). Positions with no gnomAD variant entry or fewer than 50 heteroplasmic carriers receive $y_{\text{het}} = -1$ and are excluded from the het loss. This trains the model to predict population-level heteroplasmy prevalence at known variant positions; per-individual allele-fraction prediction, which requires individual-level sequencing data, is not demonstrated in this preprint.

Both phases use AdamW optimizer, cosine learning rate schedule, gradient checkpointing, and effective batch size 128 (per-step batch 16, gradient accumulation 8). Phase 1: peak LR 1×10^{-4} , 2,000-step warmup, 50,000 steps. Phase 2: peak LR 3×10^{-5} , 500-step warmup, 25,000 steps, with a fresh optimizer state for domain-adaptive pre-training. Phase 1 training converged to a final MLM validation loss of 4.25 (perplexity 70), well below the uniform-prior random baseline of 8.32 (perplexity $\approx 4,087$ over the 4,102-token vocabulary), confirming that the encoder learned non-trivial sequence representations prior to any task-specific evaluation.

3.3 Evaluation Setup

Haplogroup zero-shot evaluation. We retrieved 15,741 NCBI human mitochondrial genome records with haplogroup annotations (Entrez query: `human[Organism] AND mitochondrion[Filter] AND complete genome[Title] AND haplogroup[Title]`), retaining 13,884 sequences with unambiguous major-class assignments across 26 PhyloTree groups. A stratified 80/10/10 split (`random.state=42`) produced a train library of 1,509 sequences and a balanced test set of 757 sequences (up to 40 per class; some rare classes have fewer, minimum 4). HmtDB was inaccessible at evaluation time, so NCBI sequences were used; these overlap with Phase 1 but not Phase 2 pre-training (see Limitations). No haplogroup labels were used during pre-training.

The zero-shot k-NN uses cosine distance on mean-pooled CLS embeddings across 65 overlapping 512-token windows. All windows use circular indexing: the final window spans positions 16,384–16,568 and 0–326 in a

single forward pass, so positions 0 and 16,568 are co-attended at inference, directly exercising the circular PE (Section 3.1.1).

Pathogenicity evaluation. We use 118 ClinVar pathogenic mtDNA SNPs (review status “Pathogenic”, “Likely pathogenic”, or “Pathogenic/Likely pathogenic”; entries with conflicting interpretations of pathogenicity were excluded) and 419 gnomAD v3.1 chrM common variants ($AF > 0.01$) as the benign set. Each variant is embedded as the final-layer hidden state of the 6-mer token *starting at* the 0-indexed variant position p (covering positions p to $p+5$ in the variant-mutated rCRS window); with stride-1 tokenization, this token is uniquely identified by `position_id = p`. The input window is constructed from the rCRS reference sequence with the alternate allele substituted at position p ; the 512-nucleotide window is centred on p with circular boundary handling (positions beyond 16,569 wrap to the genome start), consistent with the windowing used during pre-training. A 5-fold stratified cross-validated cosine 5-nearest-neighbour classifier is applied: within each fold, the classifier is fit on the 80% training split and the posterior probability (fraction of 5 nearest neighbours that are pathogenic) is recorded for out-of-fold test variants. AUROC and AUPRC are computed from the pooled out-of-fold scores across all 537 variants. Confidence intervals are 1,000-replicate non-parametric bootstrap.

4 Results

4.1 Haplogroup Classification

Zero-shot 5-nearest-neighbour classification on MTDNA-FM embeddings achieves **37.9%** accuracy on 26-class haplogroup prediction (95% CI 34.4–41.2%; random baseline 3.85%; 9.9 $\times$ lift; macro-F1 0.3703; Table 1). The balanced test set contains 757 sequences across 26 PhyloTree classes drawn from 13,884 NCBI human mitochondrial genomes; the train library contains 1,509 sequences. Note that the evaluation uses a balanced test set (up to 40 sequences per class); accuracy on the natural haplogroup distribution, where haplogroup H alone comprises $\sim 14\%$ of sequences, has not been evaluated.

Figure 3 shows the UMAP of 100 sequences sampled across 50 sub-haplogroups. African root L clades, Asian M clade, and West Eurasian haplogroups (H, J, U, and relatives) form distinct regions without supervision. Misclassifications appear to be phylogenetically structured: H is confused with its ancestor HV; L0 is confused with its sibling L1. The UMAP visualization suggests that cross-clade errors (e.g., African L predicted as West Eurasian H) are rare, as major clades occupy distinct embedding-space regions; formal verification of cross-clade error rates is deferred to the extended paper.

A supervised 6-mer frequency logistic regression on the same 1,509-sequence library achieves 78.7% accuracy (Macro-F1 0.678; Table 1). This confirms the task is solvable from sequence-level k-mer statistics. The zero-shot gap reflects representation capacity, not task difficulty.

Table 1: 26-class haplogroup classification on the same 1,509-sequence train library and 757-sequence balanced test set. Zero-shot methods use frozen embeddings with cosine 5-NN; no haplogroup labels were used in pre-training. MTDNA-FM used 65 overlapping windows (full genome); DNABERT-2 used a single 3,000 nt window (18% of the genome). The 6-mer logistic regression uses the same 1,509-sequence library as the zero-shot methods.

Method	Setting	Accuracy (%)	Macro-F1
Random baseline	26-class	3.85	0.038
MTDNA-FM (zero-shot 5-NN)	26-class	37.9	0.3703
DNABERT-2 (zero-shot 5-NN)	26-class	66.3	0.6593
6-mer LR (supervised)	26-class	78.7	0.678

4.2 Variant Pathogenicity Prediction

A 5-fold stratified cross-validated cosine 5-nearest-neighbour classifier on variant-position token embeddings achieves AUROC 0.777 (95% CI 0.731–0.821) against 118 ClinVar pathogenic mtDNA SNPs and 419 gnomAD

Figure 3: Haplogroup embedding structure and 26-class classification

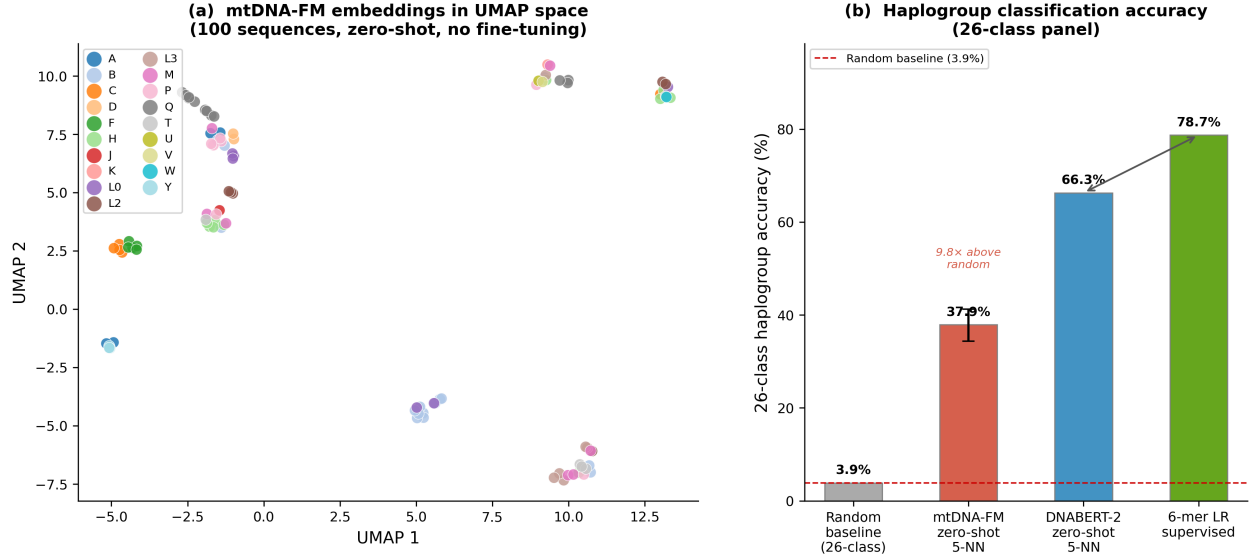


Figure 3: Haplogroup embedding structure and 26-class classification. (a) UMAP of MTDNA-FM embeddings for 100 sequences across 50 *sub*-haplogroups (used for visualization only; the classification task uses 26 major PhyloTree classes), coloured by major haplogroup. African L clades, Asian M, and West Eurasian haplogroups (H, J, U, and relatives) form separate regions without supervision. (b) 26-class haplogroup accuracy across four methods on the same 1,509/757 train/test split. Random baseline 3.85%; MTDNA-FM zero-shot 37.9% (9.9 $\times$ above random; 95% CI shown); DNABERT-2 zero-shot 66.3%; supervised 6-mer logistic regression 78.7%. Neither MTDNA-FM nor DNABERT-2 used haplogroup labels.

high-frequency benign variants (AUPR 0.440 vs. 0.220 random; Figure 4). The operating point is TPR 0.69 at FPR 0.21. No pathogenicity labels were used in pre-training.

Per-class AUROC: missense ($n_+=56$, AUROC 0.727, AUPR 0.303 vs. 0.184 random; 2.0 $\times$ lift), tRNA ($n_+=44$, AUROC 0.718, AUPR 0.773), rRNA ($n_+=5$, AUROC 0.639, limited by sample size). For tRNA variants, the class-specific random AUPR is $44/65 = 0.677$ because pathogenic tRNA variants are 68% of the tRNA subset; the observed AUPR 0.773 is only 1.14 $\times$ above this class-specific baseline, the smallest relative gain among evaluated classes. The overall 2.0 $\times$ AUPR lift primarily reflects the missense class.

The D-loop class has a single ClinVar pathogenic variant ($n_+ = 1$), which ranked below the median benign variant; AUROC is not reported for a single positive. The “other” variant class ($n_+=12$) showed below-chance performance (AUROC 0.20); with only 12 positives the bootstrap confidence interval spans nearly the full [0,1] range, so this most likely reflects sampling noise.

The signal in missense and tRNA classes likely reflects evolutionary constraint learned in Phase 1: pathogenic variants concentrate at positions conserved across vertebrates, producing distinct embeddings relative to population-common variants. Benchmarking against established variant effect predictors (SIFT, PolyPhen-2, MitoTIP, APOGEE2) is deferred to the extended paper.

5 Analysis: Four Observations from the Performance Comparison

The per-class results support four observations about how domain-specialized and generalist representations differ. Each identifies a specific cause and rules out simpler explanations. Mechanisms that require ablation to confirm are noted.

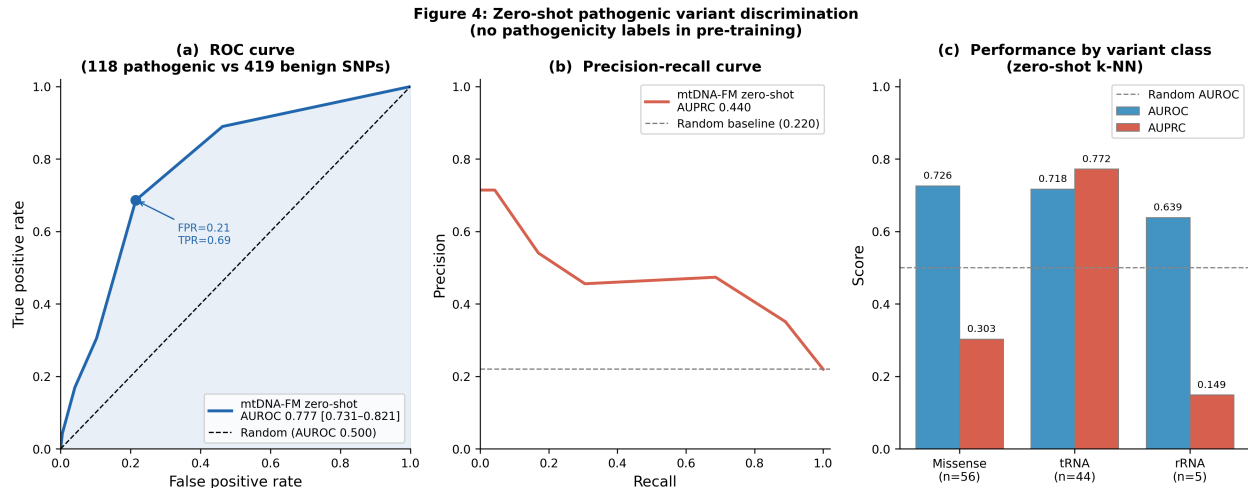


Figure 4: **Pathogenic variant discrimination (no pre-training labels)**. (a) ROC curve for 5-fold stratified CV cosine 5-NN on 118 ClinVar pathogenic vs. 419 gnomAD benign mtDNA SNPs. AUROC 0.777 (95% CI 0.731–0.821); operating point at FPR 0.21, TPR 0.69 is marked. (b) Precision-recall curve; AUPRC 0.440 vs. random baseline 0.220. (c) AUROC and AUPRC by variant class: missense (AUROC 0.727), tRNA (AUROC 0.718, AUPRC 0.773; class-specific random AUPR = 0.677), rRNA (AUROC 0.639). No pathogenicity labels were used in pre-training.

5.1 Observation 1: The Performance Gap Is Clade-Specific

Within MTDNA-FM, all nine West Eurasian haplogroups (H, HV, J, K, T, U, V, W, X) fall in F1 range 0.000–0.241. All six African L haplogroups (L0–L5) fall in 0.438–0.857 (see Table footnote for L5 caveat). These ranges do not overlap: the best West Eurasian haplogroup (U, F1 = 0.241) is strictly below the worst African L haplogroup (L3, F1 = 0.438). Across 15 independent class measurements, complete non-overlap establishes a structural failure specific to West Eurasian haplogroups in MTDNA-FM.

DNABERT-2 shows no analogous separation. Its West Eurasian range (0.400–0.957) and African L range (0.513–1.000) overlap substantially (Table 2).

If the 28.4 percentage-point overall gap were driven purely by scale [Kaplan et al., 2020], performance would improve roughly uniformly across all haplogroups. Instead, the ratio of African L to West Eurasian average F1 is $7.2\times$ within MTDNA-FM and only $1.13\times$ within DNABERT-2. This difference in clade ratios rules out a uniform scale effect.

Table 2: Clade-level average F1 from zero-shot cosine 5-NN (same train/test split). West Eurasian haplogroups: H, HV, J, K, T, U, V, W, X. African L clade: L0–L5. Full per-class values in Supplementary Tables S1 and S3.2.

Clade	Classes	Average F1		Gap	mtDNA-FM F1 range
		MTDNA-FM	DNABERT-2		
West Eurasian	9	0.096	0.709	0.613	0.000–0.241
African L	6	0.692	0.803	0.111	0.438–0.857 ^a

^aUpper bound (0.857) is haplogroup L5, $N = 4$ test sequences; excluding L5, the African L range is 0.438–0.816. Non-overlap with West Eurasian range (0.000–0.241) holds either way.

5.2 Observation 2: mtDNA-FM’s West Eurasian Haplogroup Failure Is a Representation Failure, Not an Information-Access Failure

MTDNA-FM processes all 16,569 positions of each genome across 65 overlapping windows. DNABERT-2 was evaluated on the first 3,000 nt, covering the D-loop control region (positions 1–576), the 12S rRNA gene

(648–1601), and the first 1,329 nt of the 16S rRNA gene (1671–3000). The ND1 coding gene does not begin until position 3,307 and falls outside DNABERT-2’s window.

Key diagnostic positions for the worst-performing West Eurasian haplogroups fall within the region DNABERT-2 covers. Haplogroup H is partially defined by positions 750 and 2706 [van Oven and Kayser, 2009], both within 3,000 nt. Haplogroup J carries its defining D-loop variant at position 295 [van Oven and Kayser, 2009]. Using only these early markers, DNABERT-2 achieves $F1 = 0.643$ on H and $F1 = 0.947$ on J.

MTDNA-FM covers the same positions (within the first ~ 11 of its 65 windows, stride 256, window 512) yet achieves $F1 = 0.055$ on H and $F1 = 0.149$ on J. The discriminative information is in positions 1–3,000 nt. DNABERT-2 uses it. MTDNA-FM does not. The failure is therefore in how MTDNA-FM represents what it sees, not in what it has access to.

Three candidate mechanisms are consistent with this observation; each requires ablation to isolate:

(a) Scale. At 5.8M vs 117M parameters, MTDNA-FM has a $20\times$ smaller parameter budget. This reduces representational capacity for subtle sequence differences between closely related haplogroups.

(b) Tokenization. Stride-1 6-mer tokenization creates approximately 16,569 tokens per genome with extreme inter-token autocorrelation: tokens at positions i and $i + 1$ share five of six nucleotides. DNABERT-2’s BPE vocabulary [Zhou et al., 2024] of approximately 500 non-redundant tokens provides higher information density per token, particularly for haplogroup-diagnostic sequence patterns.

(c) Window mean-pooling. MTDNA-FM uses 65 overlapping 512-token windows with stride 256, producing a single 256-dimensional vector from the mean of 65 window CLS representations. A diagnostic signal at D-loop position 295 contributes equally to a conserved tRNA motif at position 7,029. This dilutes position-specific markers to roughly $1/65$ of the total embedding.

5.3 Observation 3: Bidirectional Precision and Recall Collapse Confirms Embedding Overlap Within the West Eurasian Haplogroup Group

For a k-NN classifier, low recall on class H means H test sequences are assigned to other classes. Low precision means non-H sequences are assigned to H. Both failing simultaneously means H embeddings occupy the same region of cosine space as embeddings from other haplogroups: the k-NN cannot distinguish them.

Table 3 reports precision and recall for West Eurasian haplogroups in MTDNA-FM. Eight of nine show both precision and recall below 0.20 (the sole exception is U, with precision = 0.389). For comparison, haplogroup L0 has precision = 0.857 and recall = 0.750: its embeddings form a distinct cluster. The UMAP in Figure 3 shows this directly: haplogroup H is confused with its ancestor HV, consistent with near-zero F1 for both.

Table 3: Precision and recall for West Eurasian haplogroups under MTDNA-FM zero-shot 5-NN. Low precision and low recall together indicate bidirectional embedding overlap: H sequences are assigned to other classes (low recall) and non-H sequences are assigned to H (low precision). African root clade L0 (precision 0.857, recall 0.750) is shown for comparison.

Haplogroup	Precision	Recall	Support
H	0.061	0.050	40
HV	0.036	0.040	25
J	0.185	0.125	40
K	0.182	0.150	40
T	0.133	0.050	40
U	0.389	0.175	40
V	0.056	0.091	11
W	0.000	0.000	16
X	0.071	0.083	12

5.4 Observation 4: Full-Genome Coverage Provides Measurable Advantage Where DNABERT-2 Is Truncated

Three haplogroups show MTDNA-FM outperforming DNABERT-2: C (0.632 vs 0.611), F (0.716 vs 0.613), and E (0.400 vs 0.222). These haplogroups have PhyloTree-documented diagnostic positions beyond position 3,000 nt [van Oven and Kayser, 2009]. DNABERT-2’s single-window evaluation truncates each genome to the first 3,000 nt; MTDNA-FM’s 65-window protocol accesses the full genome including these positions.

The result is most reliably supported by haplogroups C ($N = 40$) and F ($N = 36$). Haplogroup E has only 4 test sequences and its result (F1 0.400 vs. 0.222, based on 1–2 additional correct predictions) is anecdotal. This is an evaluation-protocol observation: it is specific to DNABERT-2’s single-window evaluation and does not extend to a full-genome DNABERT-2 protocol, which has not been tested.

5.5 Design Implications for Next-Generation mtDNA Models

Each observation identifies a specific improvement target.

Observations 1 and 3 (clade-specific failure and embedding overlap) point to representation quality for recently diverged, sparse-signal clades. Scaling to 100M parameters is the highest-leverage single intervention [Kaplan et al., 2020]. Balanced sampling during Phase 2 pre-training, with haplogroup-uniform weighting, is a lower-cost intervention that directly targets the 14.3% H over-representation in HmtDB.

Observation 2 (representation failure) points to tokenization and aggregation. BPE tokenization trained on human mtDNA sequences would reduce autocorrelated 6-mer tokens to approximately 500 non-redundant tokens [Zhou et al., 2024] and enable a single-window forward pass. Hierarchical attention over window CLS representations would let the model weight windows by diagnostic content, ending the $\sim 1/65$ dilution of D-loop signal.

Observation 4 (full-genome advantage under the evaluation protocol used) establishes that full-genome coverage provides measurable advantage when diagnostic positions fall outside a single-window truncation limit. BPE tokenization of the full 16,569 bp genome (approximately 500 tokens) would eliminate windowing entirely, resolving the aggregation problem while keeping full-genome access.

The circular positional encoding addresses the circular topology of mtDNA, a property not handled by any existing genomic model. The heteroplasmy projection channel is an architectural design that enables future work on population-level and per-individual allele-fraction modeling; its empirical contribution is not evaluated in this preprint. A scaled successor should preserve both.

6 Discussion

MTDNA-FM shows that a compact 6M-parameter encoder pre-trained on mitochondrial sequences encodes evolutionary structure without supervision. Haplogroup identity is recoverable from frozen embeddings at 37.9% 26-class accuracy. Pathogenic variants are discriminated from population-common variants without labels (AUROC 0.777). The four observations in Section 5 identify specific causes of the gap with DNABERT-2 and point to concrete improvements.

Circular PE generalizability. The circular positional encoding is not specific to mtDNA. Any circular DNA molecule benefits from removing the artificial positional discontinuity at the sequence junction.

Prokaryotic genomes and synthetic biology. Bacterial chromosomes (*E. coli*: 4.6 Mb; *B. subtilis*: 4.2 Mb) and all plasmids are circular. A foundation model with circular PE would correctly represent regulatory elements that span the origin of replication.

Extrachromosomal DNA (ecDNA) in cancer. ecDNA occurs in approximately 14% of human cancers [Turner et al., 2017] as circular elements (0.1–10 Mb) carrying amplified oncogenes and super-enhancers. Its circular topology enables long-range enhancer contacts and rapid copy-number amplification [Kim et al., 2020]. A model with circular PE would correctly represent regulatory grammar at the oncogene–enhancer junction.

Minicircle DNA vectors. Minicircle vectors are small (2–5 kb) circular constructs used in gene therapy [Kay et al., 2010]. The expression cassette forms a closed circle. Circular PE would correctly represent regulatory elements that are physically adjacent across the vector junction.

Ring chromosomes. Ring chromosomes arise from telomeric end fusion into a closed circle. They are associated with clinical syndromes (Ring 14 epilepsy, Ring 20 nocturnal epilepsy, Ring 22 intellectual disability)

[Guilherme et al., 2011] and appear in cancer cytogenetics. Circular PE accurately represents the physical topology that linear PE misrepresents at the ring junction.

We provide the circular PE implementation as a reusable `nn.Module` with a documented API.

Why does biologically correct positional encoding not overcome the performance gap? Three effects explain this. First, the $20\times$ parameter gap (5.8M vs. 117M) dominates: a model without capacity to encode fine-grained haplogroup-discriminating features cannot be rescued by PE alone. Second, the D-loop region (positions 16,024–576, where circular PE matters most) is captured by both models. The critical failure in Observation 2 is in representing diagnostic variants within the first 3,000 nt, not at the junction boundary. Third, under this evaluation protocol (single 3,000 nt window for DNABERT-2; 65-window full-genome for MTDNA-FM), MTDNA-FM outperforms on haplogroups C, F, and E, whose defining markers fall beyond DNABERT-2’s truncation limit. A full-genome DNABERT-2 evaluation has not been tested.

Circular PE likely provides a latent advantage at the D-loop junction, currently masked by the parameter budget gap. A controlled ablation comparing circular vs. sinusoidal PE on an otherwise identical model will isolate this contribution in the extended paper. At 5.8M parameters, scale is the dominant variable.

Limitations.

- *Phase 1 checkpoint only.* Both evaluations use the Phase 1 checkpoint. Phase 2 training (34,975 HmtDB sequences, heteroplasmy loss) was executed but not evaluated for this preprint due to compute constraints. Phase 1 vs. Phase 2 ablation is deferred to the extended paper.
- *Evaluation–pre-training overlap.* The haplogroup evaluation uses NCBI sequences, which also contributed to Phase 1 pre-training. No haplogroup labels were used in pre-training, but some evaluation sequences may have been seen during MLM training. The fraction of the 757-sequence test set that appeared in Phase 1 pre-training has not been estimated. If substantial, the 37.9% result may partly reflect sequence-level memorisation. DNABERT-2 had no such overlap, so the 28.4 percentage-point gap should be treated as a lower bound on DNABERT-2’s advantage; the true gap on fully held-out sequences could be larger.
- *European cohort bias.* HmtDB contains approximately 60–70% European-origin sequences; haplogroup H alone is approximately 14.3% of labeled sequences. Whether this over-representation during Phase 2 contributes to embedding-space overlap for West Eurasian haplogroups is a testable hypothesis (Section 5.1).
- *SNPs only.* The pathogenicity and heteroplasmy models handle single-nucleotide variants. Insertions and deletions require a different tokenization strategy. The AF >1% benign proxy may include haplogroup-defining positions that are fixed at high frequency in specific populations; haplogroup-frequency confounding has not been assessed.
- *No wet-lab validation.* All results are computational. Experimental validation of predicted pathogenic variants is future work.
- *Haplogroup resolution.* Sub-haplogroup prediction (H1a vs. H1b) requires finer-grained labels not included in this study.

Future work. The extended paper will include ablations of circular PE vs. sinusoidal PE and the heteroplasmy channel, a Phase 1 vs Phase 2 curriculum ablation, supervised fine-tuning, and a pathogenicity benchmark against MitoTIP and APOGEE2.

7 Conclusion

MTDNA-FM is the first foundation model dedicated to mitochondrial DNA. Its circular positional encoding correctly represents the circular topology of the genome at the D-loop junction, a property not addressed by any existing genomic model. Zero-shot evaluation confirms that the Phase 1 checkpoint encodes evolutionary structure without supervision: 37.9% on 26-class haplogroup classification ($9.9\times$ the random baseline) and AUROC 0.777 on pathogenic variant discrimination.

Comparison with DNABERT-2 reveals a 28.4 percentage-point gap that is clade-specific and traceable to representation quality, not information access. All nine West Eurasian haplogroups fall in F1 range 0.000–0.241; all six African L haplogroups fall in 0.438–0.857, with no overlap. DNABERT-2 correctly classifies H and J from diagnostic positions that mtDNA-FM also covers, indicating a representation failure rather than a coverage failure. The four observations in Section 5 map four concrete targets for a next-generation model: scaling to 100M parameters, BPE tokenization to reduce token autocorrelation, hierarchical window aggregation, and haplogroup-balanced pre-training. The circular PE and heteroplasmy channel should be preserved; they address structural properties of the genome with no counterpart in existing models.

All code, model weights, and the DVC pipeline are publicly available.

Data Availability

Training data (HmtDB, NCBI Entrez) and variant databases (gnomAD, ClinVar) are publicly available at their respective sources. No proprietary or restricted data were used. Pre-trained model weights are available at <https://huggingface.co/vthawfeek/mtdna-foundation-model>.

Code Availability

All code is available at <https://github.com/vthawfeek/mtdna-foundation-model> under the Apache License 2.0. The full DVC pipeline for end-to-end reproducibility is included.

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Competing Interests

The author declares no competing interests.

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⁴⁷⁷ **Supplementary Material**

⁴⁷⁸ *See supplementary.tex for detailed hyperparameter tables, full per-class metrics, ablation learning curves, and*
⁴⁷⁹ *mathematical proofs.*